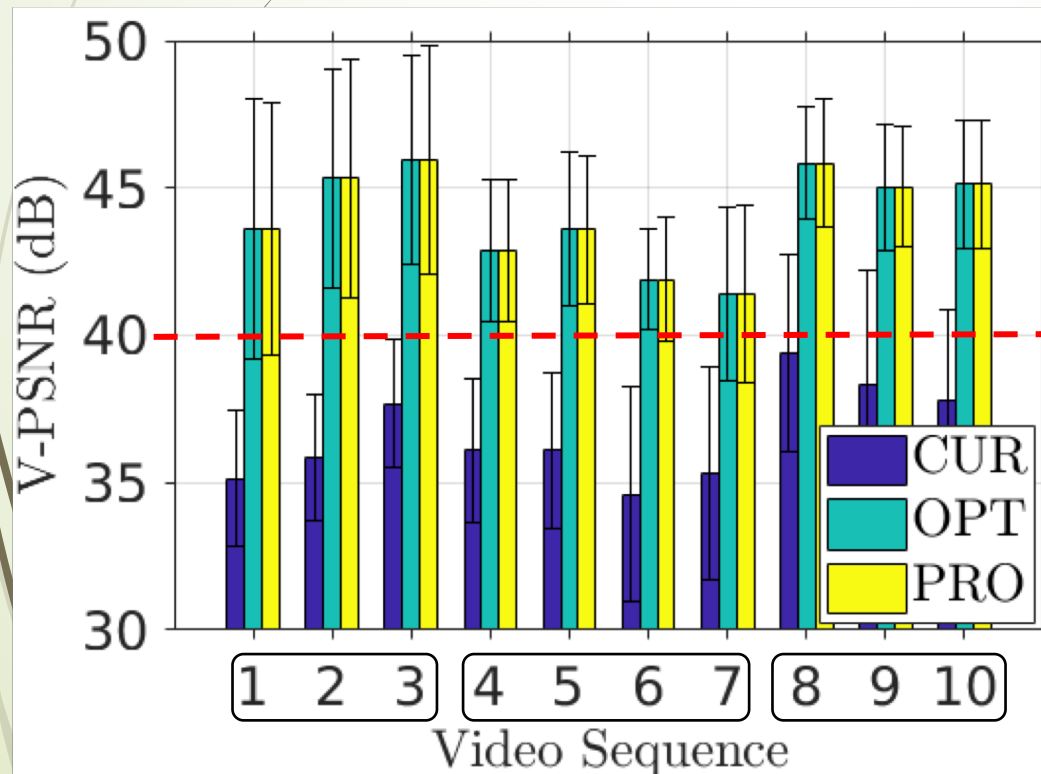


Simulation Setup

- Hardware
 - Edge server: Intel 40-cores workstation with 256 GB RAM
 - HMD client: Oculus Rift DK2 with an i7 PC
- Encoding
 - No. tiles = $\{5 \times 5\}$
 - DASH segment length = $\{2\}$ secs
 - Encoding bitrates outside/inside viewport = $\{1, 8\}$ Mbps
 - FoV size = $\{100^\circ \times 100^\circ\}$
- Viewers: Randomly select 40 traces from the dataset (40/100)
- Network: The outbound bandwidth of the edge server is set to 1 Gbps
- Baselines
 - Current streaming approach (**CUR**)
 - CPLEX Solver (**OPT**)
 - Proposed system (**PRO**)

Sample Results: Better Video Quality

- High video quality (V-PSNR ≥ 40 dB)
- The min/avg/max improvements are 6/7.4/8.4 dB



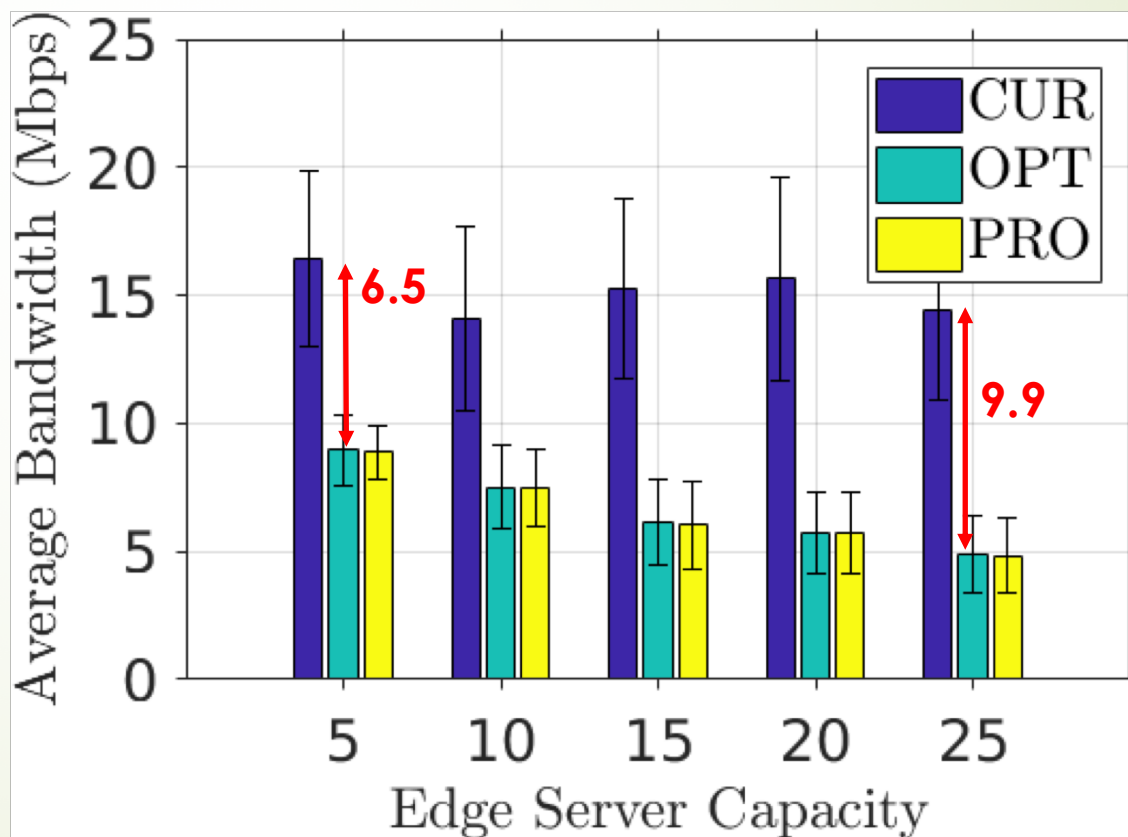
{1, 2, 3}: Natural, Fast

{4, 5, 6, 7}: Natural, Slow

{8, 9, 10}: Synthesis, Fast

Sample Results: Reduced Bandwidth Consumption

- Save min/avg/max 35%/56%/62% bandwidth consumption
- Higher edge capacity, more bandwidth



Agenda

- Cameras and Displays
- Projection Models
- Tiled Streaming (PA1)
- Edge Rendering
- Fixation Prediction
- Other Challenges

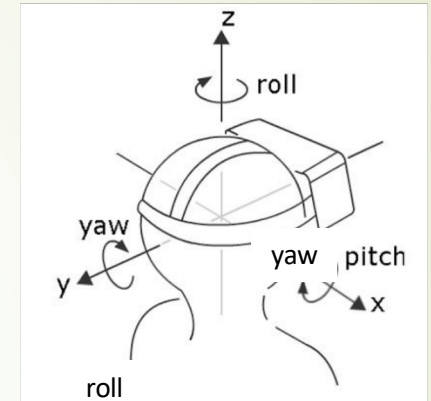
Predict the Viewers' Head Movements

- Q: Why bother to predict that?
- A: Want to know which FoV should we stream to meet the viewer's needs in the next moment (a few secs from now)
- Goal: predict which tiles are most likely viewed by the viewers
 - ➔ which tiles should be included in the next segment

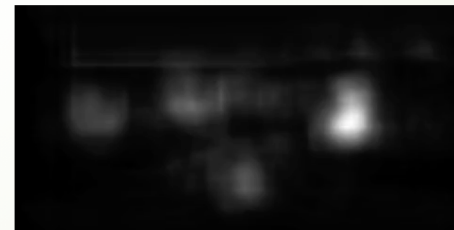
69.7%	64.3%	67.2%	0.7%
80.2%	91.2%	11.2%	0.2%
23.1%	10.2%	2.1%	0.1%
0.7%	1.4%	0.3%	0.5%

New Way to Predict Fixations

- Neural network with the features
 - **content**: saliency and motion maps
 - **sensor**: ← new, only from HMDs viewer's yaw, roll, and pitch



saliency
detection



motion
detection

